

Information entropy and NLP applied to Musicology

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Backgrounds

- Traditional researches on musicology are mainly qualitative
- Applying AI at music analysis & creation
- Inspiration from Information theory (entropy) and NLP technologies (N-gram)

Harmonic Entropy

$$H(\text{Chords}) = \sum_{\text{chord} \in \text{Chords}} p(\text{chord}) \log \frac{1}{p(\text{chord})}$$

- Dm7 – G7 – C: Typical IIm⁷ – V⁷ – I progression in C major
- Unigram model: $P(\text{Dm7} \rightarrow \text{G7} \rightarrow \text{C}) = P(\text{Dm7})P(\text{G7})P(\text{C})$
- Bigram model: $P(\text{Dm7} \rightarrow \text{G7} \rightarrow \text{C}) = P(\text{Dm7})P(\text{G7}|\text{Dm7})P(\text{C}|\text{G7})$
- Trigram model: $P(\text{Dm7} \rightarrow \text{G7} \rightarrow \text{C}) = P(\text{Dm7})P(\text{G7}|\text{Dm7})P(\text{C}|\text{Dm7} \rightarrow \text{G7})$

$$P(w_{1:N}) = \prod_{j=1}^N P(w_j | w_{1:j-1}) = \prod_{j=1}^N P(w_j | w_{j-n+1:j-1})$$

Research Goals

- Calculate harmonic entropy of classical music literatures
 - J. S. Bach: Prelude and Fugue in C major, BWV 846 (a)
 - L. v. Beethoven: Piano Sonata No. 8, Op. 13. II: Adagio cantabile (b)
 - L. v. Beethoven: Piano Sonata No. 8, Op. 13. III: Allegro (c)
 - F. Chopin: Étude Op. 10, No. 1 (d)
- Build a model that predicts chord progression

Methodology

- Building dataset
 - Annotate chord progression of each piece in .txt format
 - Consider only basic triads and sevenths (treat augmented 6ths as dominant 7ths, ignore pedal tones, etc.)
 - Format: [root][quality]/[inversion]
 - Start and end of phrases notated as a dot

C Dm⁷/C G⁷/B C

C/0
Dm7/3
G7/1
C/0

I ii⁴ V⁵ I

- Calculate harmonic entropy
 - Case 1: Treat inverted chords differently
 - Case 2: ignore inversions
- Generate chord progression
 - Backoff model
 - Decrease n-gram to (n-1)-gram when the former works poorly
 - Trigram → Bigram → Unigram

Experiments

Comparing harmonic entropy

	Bach WTC	Beethoven (mov. 2)	Beethoven (mov. 3)	Chopin Étude
Inversion	4.327	4.215	4.944	4.708
noInversion	3.715	3.452	4.070	4.531

- Inversions considered, (c) shows highest harmonic entropy may due to frequent modulations (Cm → Eb → Cm → Ab → Cm → C → Cm)
- Inversions ignored, (d) shows highest harmonic entropy may due to increased chromaticism

Predicting next chord (major key)

Bigram				Trigram			
Given	C/E	D ⁷	G	Given	Dm ⁷ - G ⁷	C/G - G ⁷	F - F#dim ⁷
Predicted	A ⁷	G G ⁷ sus4	G ⁷	Predicted	C	C	G

Generating chord progression

Chord progressions starting from C (I)

Imaj	V7/3	Imaj/1	V7/1	Imaj	Vmaj/1	VIm	II7/2
Imaj	.	V7	Imaj	.	V7	Imaj	.
Imaj	.	Imaj	IVmaj/1	#IVm7b5/1	Vsus4	Vmaj	Imaj
Imaj	IVmaj	#IVm7b5	Vmaj	II7	V7sus4	V7	Imaj

Chord progressions starting from G (V)

Vmaj	V7	Imaj	.	V7	Imaj	.	Im
Vmaj	.	#IIm7/2	IIIm/1	VIIIm7/2	Imaj/1	.	IVmaj7/3
Vmaj	.	II7	V7sus4	V7	I7	IVmaj/2	IIIm7
Vmaj	.	VIIIm7b5/2	V7/3	Imaj/1	VI7	IIIm	V7

9 C F F#m⁷(b5) G D⁷ G⁷(sus4) G⁷ C

Predicted

resolutions of
(secondary) dominants
and diminished 7ths

Concluding Remarks

- Propose a quantitative method to analyze harmonic complexity
 - Large corpus is necessary to derive significant results
- Build a chord generation model
 - Model identifies simple progressions, but fails to learn longer contexts